



Porta Ignition QC Process Control System

Complete Project Framework & Implementation Blueprint

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1. Project Intent & Executive Summary

1.1 Intent

The primary objective of this project is to completely replace controlled paper logs and manual Excel spreadsheets across the Myrtleford mill floor with authenticated, station-aware Ignition Perspective web forms backed by a high-performance central relational SQL database. The application incorporates automated statistical tolerance validations, managed non-conformance action trees, and real-time process monitoring dashboards to deliver centralized quality assurance.

1.2 Executive Summary

This platform is architected as an extensible, configuration-driven plant-wide electronic process-control engine, moving away from standalone, static application pages. A unified, reusable application shell dynamically loads the active target form based on operational station context, resolves effective-dated engineering specifications, enforces authorized user signatures, tracks numeric tolerances live, triggers immediate floor recheck workflows, and maintains an unalterable system audit trail.

The application boundary focuses squarely on a two-workspace paradigm (Operational Station Logs and Management Analytics) within a single Ignition Perspective project using optimized, shared backend scripting objects. Advanced statistical process control (SPC) math engines and fully automated item-by-item product genealogy tracking systems are explicitly deferred to subsequent revision horizons, ensuring a tight, value-driven execution timeline for the immediate pilot launch.

1.3 Core Success Measures

- **Zero Paper Reliance:** Total elimination of paper logs at any commissioned pilot station.
- **Flawless Attribution:** Direct tracking linking every raw observation to an authenticated user ID, physical station context, operational shift, production tracking identifier, system timestamp, dynamic form revision, and applied engineering specification snapshot.
- **Immutable Enforcement:** Out-of-specification values automatically lock the user path, prompting mandatory action checklists without allowing operators to override the system.
- **Tamper-Evident Logs:** Submitted entries are permanent; adjustments require authenticated reason-coded changes that retain historical values.
- **Rapid Form Scalability:** Deploying subsequent forms relies primarily on data configuration within the existing database schema rather than custom engineering design.

2. User Roles, Journeys, & Access Control Matrix

The system collapses complex, multi-tiered role structures into two core functional application identities, separating shop floor data capture from administrative management tools.

QC_Manager

- └─ Inherits: All QC Technician Functions
 - └─ Manager Analytics & Full Trend Views
 - └─ Plant-Wide Overview & Live Exception Queues
 - └─ Cross-Stage Tracking Timeline
 - └─ Configuration Approval & Specification Management
 - └─ Disposition, HOLD, and Process Closure Authority

2.1 User Journeys

The QC Technician Journey: The operator logs into a localized floor station terminal using corporate Identity Provider (IdP) badges. The station home screen displays active due, upcoming, and overdue inspection frequencies. Opening a form, the technician inputs observations via clear touch interfaces. The system provides instant color-coded and text feedback. If out-of-spec conditions occur, the UI locks down the standard path and opens an actionable correction sub-panel requiring immediate rechecks, fault attribution, or material quarantine logs.

The QC Manager Journey: The manager opens a global operational view providing aggregate metrics across all active mills. An prioritized Live Exception Queue highlights outstanding floor failures. The manager can click an alert to view raw measurements, equipment calibration statuses, and technician comments. They hold absolute authority over material isolation logs, releasing holds or re-routing products for scrap or rework, and adjusting configuration specifications directly within runtime parameter views.

2.2 Simplified Access Matrix

Function Scope	QC Technician	QC Manager	Technical Admin / Engineer
Enter Process-Control Data	Yes	Yes	No
Save Draft / Submit Inspection	Yes	Yes	No
Complete Mandatory Rechecks	Yes	Yes	No
View Recent Station Records & History	Yes	Yes	No
Search Historical Records Across Plant	Yes	Yes	No
Create QC Exception / Initial HOLD	Yes (Enforced)	Yes	No
Record Final Material Disposition	Limited / No	Yes	No
Release Active Product HOLD	No	Yes	No
View Live Due/Overdue Check Status	Yes	Yes	No
View Plant-Wide Compliance & Metrics	Limited	Yes	No
View Visual Trend Graphs & Analytics	Limited	Full	No
Cross-Stage Traceability Tracking	View Linked	Full Timeline	No
Manage Form Definitions & Specs	No	Yes	No
Modify Gateway, DB, or Script Code	No	No	Yes (Designer Only)

3. Recommended Application Structure & Sitemap

The platform runs inside a single centralized Ignition Perspective project, establishing unified routing rules, streamlined navigation components, and global session property states.

3.1 Unified Project Sitemap

QC PROCESS CONTROL (Root Project)

- |
- |— Home (Default Landing for Technicians)
 - |— Due Checks Banner
 - |— Open Rechecks View
 - |— Open Exceptions Queue
 - |— Recent Submissions Log
- |
- |— Process Forms (Navigation Grouped by Area)
 - |— Veneer Preparation (Lathe Outfeed, Dryer Infeed/Outfeed, Peeler Core Temp)
 - |— Veneer Assembly (Composer Infeed/Outfeed, Core Stacker)
 - |— Layup & Press Prep (Layup Line/Extruder, Layup Outfeed, Steamer Cart)
 - |— Finishing (Trim Saws, Formply Trim Saws, Sanding Line)
 - |— Tooling (Knife Setup, Knife Grinding)
- |
- |— QC Operations (Shared Tracking Workspace)
 - |— Current Shift Log
 - |— Exceptions Queue
 - |— HOLD Register
 - |— Record Search Engine
- |
- |— Manager (Restricted to QC_Manager Role)
 - |— Overview Dashboard (Plant KPIs & Status Matrix)
 - |— Cross-Stage Tracking Timeline
 - |— Process Trends & Analytics
 - |— Product Quality Profiles
 - |— Compliance Dashboard
 - |— Configuration Management (Specifications, Forms, Schedules)

4. UX Design System & Responsive Layout Engine

4.1 Corporate Palette & Design Tokens

Design Token	Target Application	HEX Code	Visual Sample
Porta Red	Brand Identity, Critical Accents, Core Branding Elements	#D7194B	
Dark Charcoal	Main Body Typography, Grid Headers, Structural Text	#343A40	
Light Grey	Application Canvas Background, Disabled Fields, Read-Only	#F3F3F3	
Mid Grey	Section Dividers, Alternating Table Rows, Secondary Borders	#A6A6A6	
Warning Yellow	Attentions, Impending Due Windows, Recheck Triggers	#FFF200	
Amber	Overdue Check Status, Intermediate Risk Alerts	#FFBF00	
Positive Green	Compliance Met, PASS Badges, Successful Submissions	#00E62E	
Review Purple	Material Holds, Verification Triggers, Pending Dispositions	#9999FF	
Failure Red	Out-of-Specification Fields, Active Exception Banners	#E53935	

4.2 Core Interaction Rules

- **Persistent Context:** The header component continuously displays station location, user identity, current shift, active specification context, and database connection integrity.
- **Touch-Target Sizing:** All primary buttons, controls, and fields maintain a minimum height of **48px to 56px** to allow reliable manipulation while wearing production gloves.
- **Inline Threshold Visibility:** Numeric entry fields display engineering units alongside the targeted engineering tolerances right next to the value input slot.
- **Physical Layout Mapping:** Where inspection fields mirror multi-zone equipment (e.g., thickness measurements across a panel width), the fields are explicitly labeled and sorted as Left-Hand Side (**LHS**), Middle (**MID**), and Right-Hand Side (**RHS**).

4.3 Responsive Grid Viewports

To support varying hardware configurations across the floor, the Perspective engine enforces structural layouts using fluid percentage containers rather than absolute pixel coordinates:

- **Desktop & Landscape Tablets:** Split-screen layout displaying the data entry form on the left (55-65% width) and the recent station shift log on the right (35-45% width).
- **Portrait Tablets:** Elements stack vertically, placing the main input workspace above a summary block rendering the last 3 to 5 history rows.

- **Handheld Terminals:** Content transitions into a tabbed navigation bar ([Current Check], [Shift Log], [Exceptions]), disabling large nested tables to prevent horizontal scrolling.

5. Component Status, Validation, & Attention Animation

5.1 Field Validation Lifecycle Model

Visual State	Presentation Standards	System Behavior & Logic
Not Entered	Neutral White/Grey Canvas	Empty field parameter; pending user data entry input.
Unsaved / Draft	Blue or Purple Accented Indicators	Value populated locally; excluded from shift compliance statistics.
Pass	Vivid Green Border & Solid Fill Badge	Input falls safely within resolved specification bounds.
Warning	Yellow/Amber Border with alert text	Value sits inside warning bands, or instrument calibration is near expiry.
Out of Spec	Solid Red Border, Pale Red Fill, Alert Text	Measurement breaks tolerance; locks submission and initiates workflow.
QC Review	Light Purple Highlight	Record locked by floor; awaiting final disposition review from management.
HOLD	Two-tone Dark Red/Purple Banner elements	Material blocked from downstream transport; requires release credentials.

5.2 Controlled Attention Animation Policy

Visual animations are strictly restricted to initial status transitions (e.g., moving from a passing state into an out-of-specification tier). The targeted component executes exactly three subtle scale and shadow pulses over a 2.25-second window, then locks into a static, high-visibility state until resolved. Continuous text flashing is banned to prevent operator fatigue.

5.3 Advanced Perspective Stylesheet Logic (styles.css)

```
/* Persistent Validation Status Styles */
.psc-qcFieldNormal { background-color: #FFFFFF; border: 1px solid #A6A6A6; color: #343A40; }
.psc-qcFieldPass { background-color: #EFFFFE; border: 2px solid #00B52A; color: #343A40; }
.psc-qcFieldWarning { background-color: #FFF7CC; border: 3px solid #FFBF00; color: #343A40; }
.psc-qcFieldFail { background-color: #FFE5E5; border: 3px solid #E53935; color: #343A40; }

/* Transient Pulse Animations (Executed exactly three times, then terminates) */
.psc-qcPulseWarning { animation: qcPulseWarningKeyframe 0.75s ease-in-out 3; }
.psc-qcPulseFail { animation: qcPulseFailKeyframe 0.75s ease-in-out 3; }

@keyframes qcPulseWarningKeyframe {
  0%, 100% { box-shadow: 0 0 0 rgba(255, 191, 0, 0); transform: scale(1.000); }
  50% { box-shadow: 0 0 6px rgba(255, 191, 0, 0.35); transform: scale(1.015); }
}
@keyframes qcPulseFailKeyframe {
  0%, 100% { box-shadow: 0 0 0 rgba(229, 57, 53, 0); transform: scale(1.000); }
  50% { box-shadow: 0 0 6px rgba(229, 57, 53, 0.35); transform: scale(1.015); }
}
@media (prefers-reduced-motion: reduce) {
  .psc-qcPulseWarning, .psc-qcPulseFail { animation: none !important; }
}
```

5.4 Perspective Evaluation Expression Binding Layout

```
if({view.custom.status} = 'FAIL',
  'psc-qcFieldFail' + if({view.custom.pulseTrigger}, ' psc-qcPulseFail', ''),
if({view.custom.status} = 'WARNING',
  'psc-qcFieldWarning' + if({view.custom.pulseTrigger}, ' psc-qcPulseWarning', ''),
  if({view.custom.status} = 'PASS',
    'psc-qcFieldPass',
    'psc-qcFieldNormal'
  )
)
)
```

6. Form Configuration & Specification Engine

6.1 Reusable Form Family Patterns

To reduce long-term maintenance, all 15 targeted production forms map directly to five core functional interface configurations: *Dimensional / Physical Layouts* (grid structures with spatial tags), *Repeated Load Logs* (parent-child batch entries), *Time / Duration Profilers* (sequential time tracking), *Tooling Lifecycles*

(device configurations and tool wear metrics), and *Attribute Inspections* (pass/fail matrices with dynamic comment triggers).

6.2 Deterministic Specification Resolution Engine

When an operator accesses a data entry page, the system runs a deterministic resolution script that queries active database entries using the following parameter hierarchy:

Active Date → Form Revision ID → Product Code → Grade Code → Nominal Thickness → Station ID

If the engine finds zero active specifications, or discovers conflicting records, the application will freeze data entry, block database submissions, and issue an administrative system alert to prevent processing against incorrect targets.

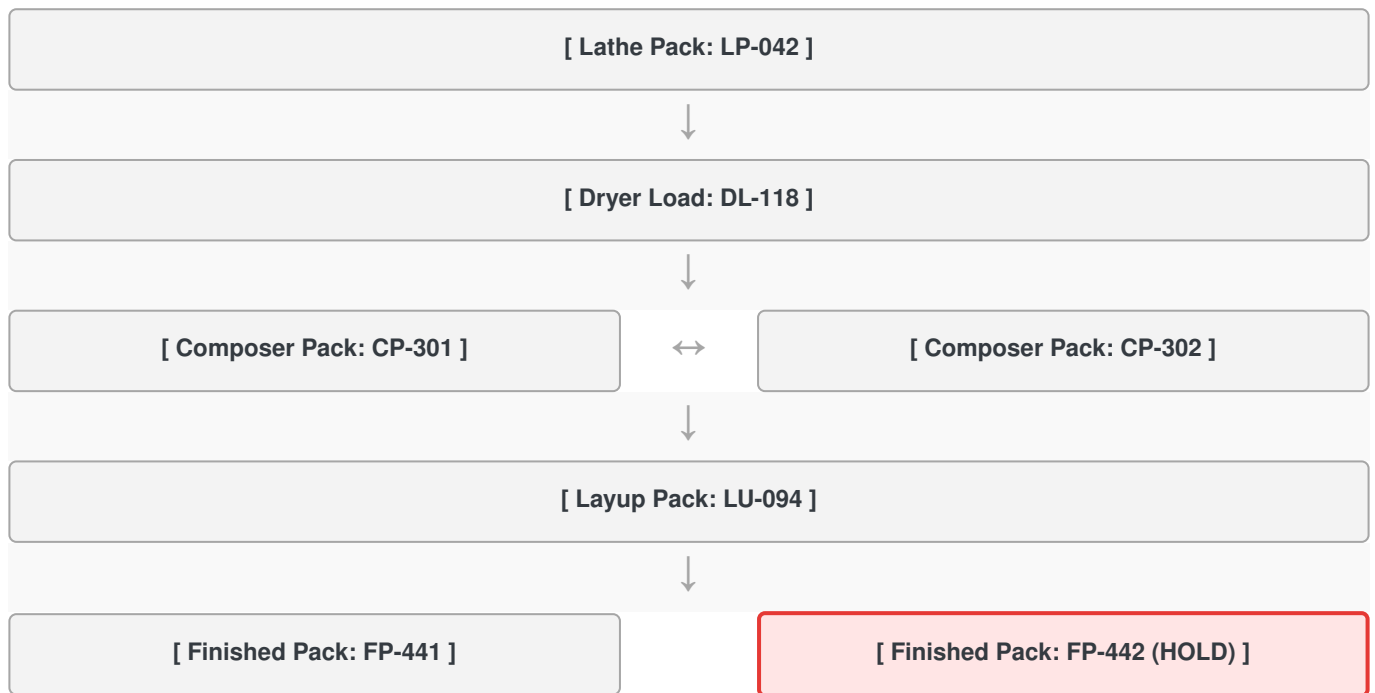
6.3 Standard Schedule Matrix

Schedule Classification	Execution Threshold Triggers	Grace Windows & Enforcement Rules
Shift Start Check	Fired during the initial 30 minutes of a new operational shift.	Mandatory 15-minute grace window; auto-escalates to Team Leader if missed.
Standard Interval	Recurring time boundaries (e.g., Hourly, Two-Hourly).	15 minutes prior to hour until 15 minutes past the hour.
Per Event / Run	Bound to material tracking signals (e.g., Pack change).	Must execute prior to finalizing the current tracking ID log.
Shift Coverage	Quantitative targets (e.g., Minimum 1 check per grade).	Validated during the final two hours of the operational shift.
Product Change	Instantly triggered by PLC changes or operator production adjustments.	Blocks subsequent batch processing until verification is submitted.
Post-Failure Escalation	Accelerated frequency rules instantiated automatically following a failure.	Mandates a check every 30 minutes until 3 consecutive passes are achieved.

7. Cross-Stage Product Tracking & Product Quality Profiles

7.1 Cross-Stage Product Tracking Architecture

Material transformations across processing areas are managed through explicit parent-child relations:



7.2 Staged Integration Deployment Strategy

- **Level 1 (Phase 4 Out-of-the-Box):** Cross-references data elements already captured by standard forms (Date, Shift, Product, Grade, Nominal Thickness, Manual Pack/Load ID, System Timestamp) to correlate entries.
- **Level 2 (Target Pilot State):** Enforces a mandatory Production Tracking ID barcode scan or manual entry at the start of every inspection check, linking database rows cleanly.
- **Level 3 (Deferred Target Model):** Integrates live tracking by linking directly to PLC tag tracking structures, automated pack label printers, and the central plant ERP database.

7.3 Cross-Stage Tracking Interface Layout

```

CROSS-STAGE TRACEABILITY ENGINE
Tracking ID: [ PT-20260716-0042 ]   Product: 17 mm EcoPly CD   Status: DISPOSITION PENDING
-----
[08:14] LATHE OUTFEED  —————> [09:08] DRYER INFEEED  —————> [11:32] DRYER OUTFEED  ————┐
  Thickness: PASS                Nominal: 3.0 mm                Moisture: OUT OF SPEC
  Width:      PASS                Actual:  PASS                Recheck:  PASS
└──────────────────────────────────────────────────────────────────────────────────────────────────┘
└──────────────────────────────────────────────────────────────────────────────────────────────────┘
▼
[13:46] COMPOSER OUTFEED ———> [15:12] LAYUP LINE      ———> [NEXT DAY 09:22] SANDING LINE
  Joint Thk: PASS                Glue Spread: WARNING                Panel Thk: OUT OF SPEC
  Bond Qual: PASS                Retest:      PASS                [!!! HOLD ACTIVE !!!]
  
```

7.4 Unified Deviations Comparison Chart

To view process trends across varying units of measurement, the dashboard offers a Normalized Deviation View calculated as follows:

Normalized Position = (Actual Value – Target Value) / Specification Tolerance

A score near zero represents tight process control, while positive or negative values track drift relative to allowable engineering limits.

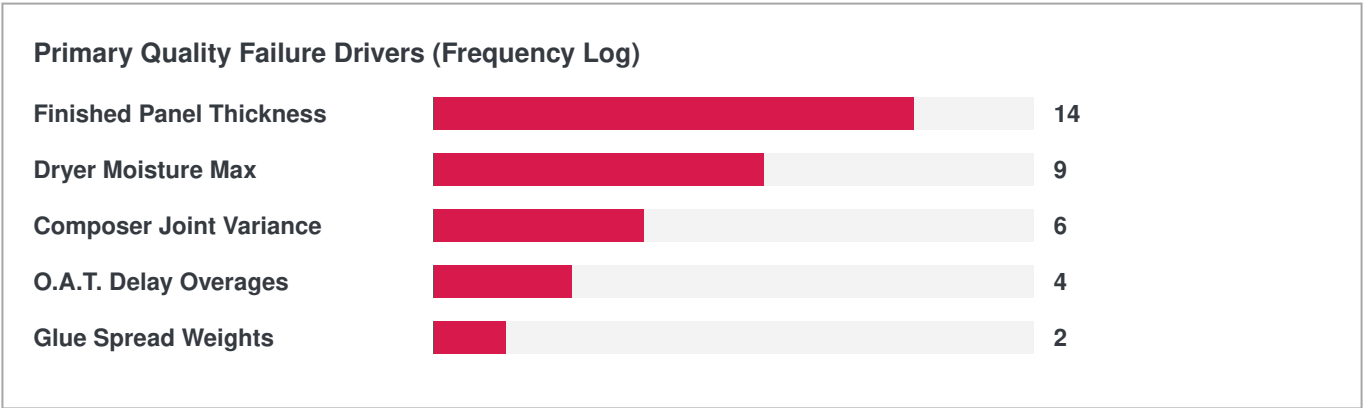
8. Quality Manager Analytics & Reporting Framework

8.1 Management Overview Dashboard Elements

The management workspace aggregates data into a real-time health matrix, tracking operational compliance, open exceptions, material insulation logs, and process control indicators across the facility.

8.2 Deep Analytics Configurations

Quality Failure Pareto Analyses: Dynamically calculates frequency counts, percentage distributions, and cumulative process impact curves to highlight the primary drivers of quality failure across the mill floor:



9. Data Architecture & Audit Model

9.1 Data Entity Schema Layout

The backend relational schema is built on strict data models to enforce integrity, track effective-dated configurations, and maintain complete system logs:

- qc_form_definition: Central registry for tracking form names, categories, and owners.
- qc_form_revision: Handles active form states (DRAFT, ACTIVE, SUPERSEDED).
- qc_specification: Relational store for min/target/max limits across products and stations.
- qc_inspection: Header record capturing timestamps, shifts, users, and aggregate status.

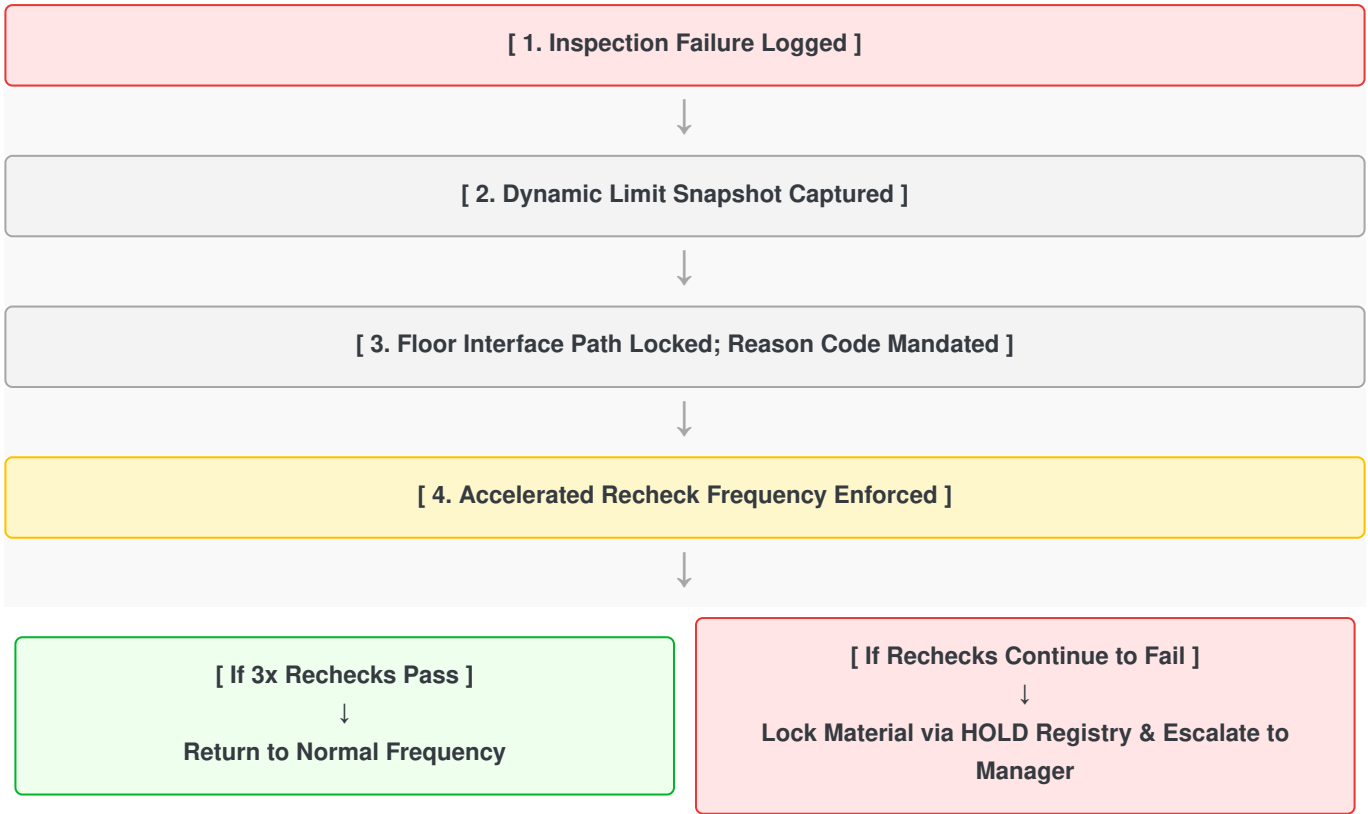
- qc_measurement: Raw observation storage mapping directly back to inspection headers.
- qc_audit_event: Immutable modification log recording user reasons for data amendments.

9.2 Data Integrity Rules

- **Limit Snapshotted Storage:** Measurement records explicitly save an absolute snapshot copy of the active specification limits enforced at the moment of submission. This decouples history from future tolerance updates.
- **Idempotence Protections:** Clients attach unique transaction tracking tokens to prevent duplicate entries caused by double-tap inputs or localized communication drops.
- **No Hard Deletions:** Data cannot be dropped from the primary logs. Corrections flag older rows as superseded while adding new records, preserving a transparent historical footprint.

10. Non-Conformance Workflows

When an observation falls out-of-specification, the system enforces a strict automated corrective workflow path, preventing user bypass and escalating structural issues to shift supervisors if conditions persist.



11. Integration, Availability, & Network Recovery

11.1 System Integration Strategy

The platform acts as a bridge between floor operations and high-level enterprise software, connecting directly to OPC automation networks for live machine parameters, ERP systems for production orders, and hardware devices like digital micrometers for direct data validation.

11.2 Offline Strategy & Network Defenses

Terminal views run active network check scripts. If the central connection drops, the interface locks down submission, transitions into a secure offline storage mode, and cues data locally. Once communication is verified stable, records sync sequentially using transactional boundaries to prevent duplicate inputs.

12. Quality Assurance, Testing, & Performance Constraints

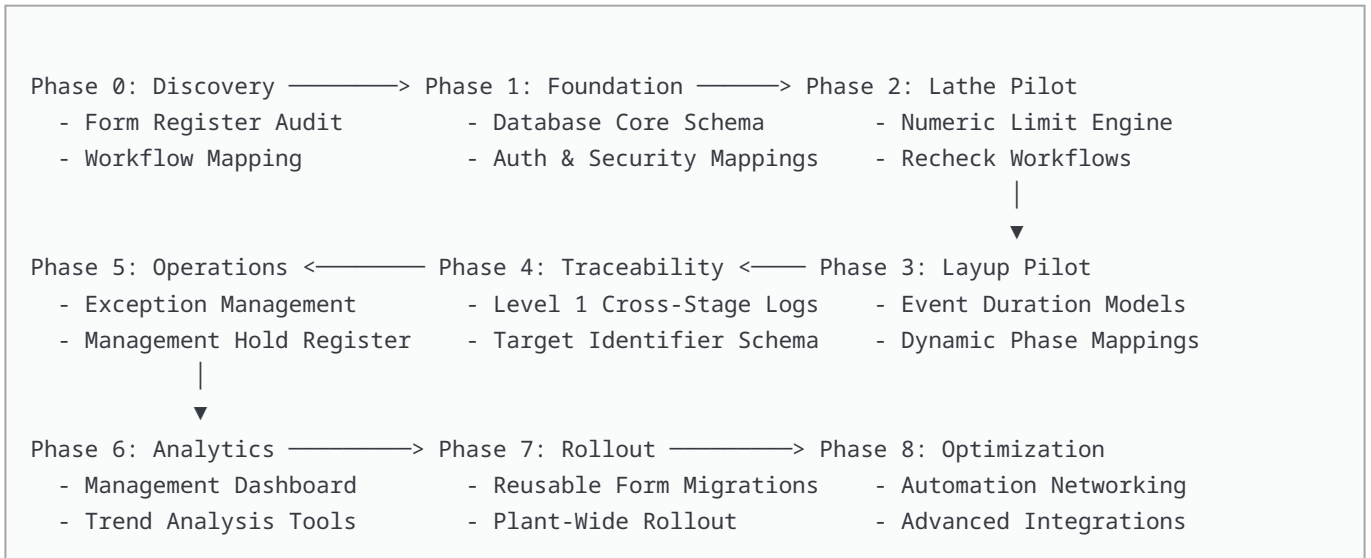
Before factory acceptance sign-off, software components undergo targeted system validation protocols. This testing matrix screens edge-case inputs (± 0.001 boundary deviations), simulates simultaneous entries to verify database lock handling, and tests layout performance on floor tablets to ensure touch target scaling standards (48px-56px) are met.

13. Initial Project Risk Register

Risk Identification	Severity	Operational Impact Summary	Framework Mitigation Architecture
Ad-Hoc Page Proliferation	High	Unique views lead to high maintenance and broken validation paths.	Enforces standard configuration layouts driven by reusable core tables.
Incorrect Spec Resolution	High	Out-of-date limits cause unrecorded defects or unnecessary stops.	Deterministic lookup hierarchy locks inputs if parameters are ambiguous.
Network Loss Data Drops	High	Dropouts cause lost history or duplicate database insertions.	Combines transactional database limits with client-side idempotency tokens.
Historical Data Distortion	High	Altering active tolerances inadvertently rewrites historical results.	Uses effective-dated records and saves absolute snapshots at submission.

14. Delivery Roadmap & Project Governance

Implementation maps across structured development blocks, beginning with database core deployment, followed by an immediate Lathe Outfeed pilot, before scaling out across remaining production centers.



15. Framework Deliverables Checklists

15.1 Component Parameter Catalog

To ensure consistent application design, all Perspective view objects follow rigid input parameter layouts, passing variables through defined components: HeaderComponent, NumericMeasurement, AttributeField, StatusBadge, and ActionRequiredPanel.

15.2 Immediate Action Items & Milestone Priorities

1. Complete the data mapping register for all remaining processing sheets.
2. Verify target database version availability and confirm Perspective system environments.
3. Secure formal sign-off for corporate design styles, verification states, and attention pulse logic.
4. Build and test core framework elements (NumericMeasurement, StatusBadge, CurrentShiftLog, ActionRequiredPanel).
5. Deploy and test the initial Lathe Outfeed form framework directly under live production floor conditions.